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Dissecting Recall of Factual Associations in Auto-Regressive Language Models

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Abstract

Transformer-based language models (LMs) are known to capture factual knowledge in their parameters. While previous work looked into *where* factual associations are stored, only little is known about *how* they are retrieved internally during inference. We investigate this question through the lens of information flow. Given a subject-relation query, we study how the model aggregates information about the subject and relation to predict the correct attribute. With interventions on attention edges, we first identify two critical points where information propagates to the prediction: one from the relation positions followed by another from the subject positions. Next, by analyzing the information at these points, we unveil a three-step internal mechanism for attribute extraction. First, the representation at the last-subject position goes through an enrichment process, driven by the early MLP sublayers, to encode many subject-related attributes. Second, information from the relation propagates to the prediction. Third, the prediction representation "queries" the enriched subject to extract the attribute. Perhaps surprisingly, this extraction is typically done via attention heads, which often encode subject-attribute mappings in their parameters. Overall, our findings introduce a comprehensive view of how factual associations are stored and extracted internally in LMs, facilitating future research on knowledge localization and editing.[1]

1 Introduction

Transformer-based language models (LMs) capture vast amounts of factual knowledge (Roberts et al., 2020; Jiang et al., 2020), which they encode in their parameters and recall during inference (Petroni et al., 2019; Cohen et al., 2023). While recent works focused on identifying *where* factual knowledge is encoded in the network (Meng et al., 2022a; Dai et al., 2022; Wallat et al., 2020), it remains unclear *how* this knowledge is extracted from the model parameters during inference.

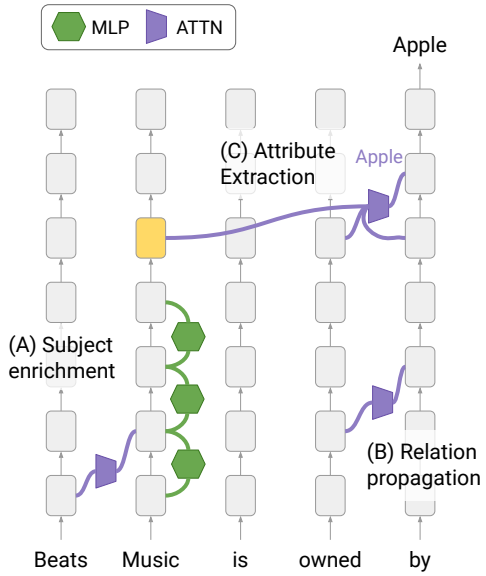

Figure 1: Illustration of our findings: given subject-relation query, a subject representation is constructed via attributes' enrichment from MLP sublayers (A), while the relation propagates to the prediction (B). The attribute is then extracted by the MHSA sublayers (C).

In this work, we investigate this question through the lens of information flow, across layers and input positions (Elhage et al., 2021). We focus on a basic information extraction setting, where a subject and a relation are given in a sentence (e.g. *"Beats Music is owned by"*), and the next token is the corresponding attribute (i.e. *"Apple"*). We restrict our analysis to cases where the model predicts the correct attribute as the next token, and set out to understand how internal representations evolve across the layers to produce the output.

Focusing on modern auto-regressive decoder-only LMs, such an extraction process could be

[1]Our code is publicly available at https://github.com/google-research/google-research/tree/master/dissecting_factual_predictions

implemented in many different ways. Informally, the model needs to "merge" the subject and relation in order to be able to extract the right attribute, and this merger can be conducted at different layers and positions. Moreover, the attribute extraction itself could be performed by either or both of the multi-head self-attention (MHSA) and MLP sublayers.

To investigate this, we take a reverse-engineering approach, inspired by common genetic analysis methods (Griffiths et al., 2005; Tymms and Kola, 2008) and the recent work by Wang et al. (2022). Namely, we artificially block, or "knock out", specific parts in the computation to observe their importance during inference. To implement this approach in LLMs, we intervene on the MHSA sublayers by blocking the last position from attending to other positions at specific layers. We identify two consecutive critical points in the computation, where representations of the relation and then the subject are incorporated into the last position: first the relation and then the subject.

Next, to identify where attribute extraction occurs, we analyze the information that propagates at these critical points and the representation construction process that precedes them. This is done through additional interventions to the MHSA and MLP sublayers and projections to the vocabulary (Dar et al., 2022; Geva et al., 2022b; Nostalgebraist, 2020). We discover an internal mechanism for attribute extraction that relies on two key components. First, a *subject enrichment process*, through which the model constructs a representation at the last subject-position that encodes many subject-related attributes. Moreover, we find that out of the three sources that build a representation (i.e., the MHSA and MLP sublayers and the input token embeddings (Mickus et al., 2022)), the early MLP sublayers are the primary source for subject enrichment.

The second component is an *attribute extraction operation* carried out by the upper MHSA sublayers. For a successful extraction, these sublayers rely on information from both the subject representation and the last position. Moreover, extraction is performed by attention heads, and our analysis shows that these heads often encode subject-attribute mappings in their parameters. We observed this extraction behavior in $\sim$70% of the predictions.

Our analysis provides a significantly improved understanding of the way factual predictions are formed. The mechanism we uncover can be intuitively described as the following three key steps (Fig. 1). First, information about the subject is enriched in the last subject token, across early layers of the model. Second, the relation is passed to the last token. Third, the last token uses the relation to extract the corresponding attribute from the subject representation, and this is done via attention head parameters. Unlike prior works on factual knowledge representation, which focus on mid-layer MLPs as the locus of information (e.g. Meng et al. (2022b)), our work highlights the key role of lower MLP sublayers and of the MHSA parameters. More generally, we make a substantial step towards increasing model transparency, introducing new research directions for knowledge localization and model editing.

2 Background and Notation

We start by providing a detailed description of the transformer inference pass, focusing on autoregressive decoder-only LMs. For brevity, bias terms and layer normalization (Ba et al., 2016) are omitted, as they are nonessential for our analysis.

A transformer-based LM first converts an input text to a sequence $t_1, ...t_N$ of N tokens. Each token t_i is then embedded as a vector $\mathbf{x}_i^0 \in \mathbb{R}^d$ using an embedding matrix $E \in \mathbb{R}^{|\mathcal{V}| \times d}$, over a vocabulary $\mathcal{V}$. The input embeddings are then transformed through a sequence of L transformer layers, each composed of a multi-head self-attention (MHSA) sublayer followed by an MLP sublayer (Vaswani et al., 2017). Formally, the representation $\mathbf{x}_i^\ell$ of token i at layer ℓ is obtained by:

$$\mathbf{x}_i^\ell = \mathbf{x}_i^{\ell-1} + \mathbf{a}_i^\ell + \mathbf{m}_i^\ell \tag{1}$$

where $\mathbf{a}_i^\ell$ and $\mathbf{m}_i^\ell$ are the outputs from the ℓ-th MHSA and MLP sublayers (see below), respectively. An output probability distribution is obtained from the final layer representations via a prediction head δ:

$$\mathbf{p}_i = \text{softmax}\big(\delta(\mathbf{x}_i^L)\big), \tag{2}$$

that projects the representation to the vocabulary space, either through multiplying by the embedding matrix (i.e., $\delta(\mathbf{x}_i^L) = E\mathbf{x}_i^L$) or by using a trained linear layer (i.e., $\delta(\mathbf{x}_i^L) = W\mathbf{x}_i^L + \mathbf{u}$ for $W \in \mathbb{R}^{|\mathcal{V}| \times d}, \mathbf{u} \in \mathbb{R}^{|\mathcal{V}|}$).

MHSA Sublayers The MHSA sublayers compute *global* updates that aggregate information from all the representations at the previous layer. The ℓ-th MHSA sublayer is defined using four

parameter matrices: three projection matrices $W_Q^\ell, W_K^\ell, W_V^\ell \in \mathbb{R}^{d\times d}$ and an output matrix $W_O^\ell \in \mathbb{R}^{d\times d}$. Following Elhage et al. (2021); Dar et al. (2022), the columns of each projection matrix and the rows of the output matrix can be split into H equal parts, corresponding to the number of attention heads $W_Q^{\ell,j}, W_K^{\ell,j}, W_V^{\ell,j} \in \mathbb{R}^{d\times \frac{d}{H}}$ and $W_O^{\ell,j} \in \mathbb{R}^{\frac{d}{H}\times d}$ for $j \in [1, H]$. This allows describing the MHSA output as a sum of matrices, each induced by a single attention head:

$$\mathbf{a}_i^\ell = \sum_{j=1}^{H} A^{\ell,j}\left(X^{\ell-1}W_V^{\ell,j}\right)W_O^{\ell,j} \tag{3}$$

$$:= \sum_{j=1}^{H} A^{\ell,j}\left(X^{\ell-1}W_{VO}^{\ell,j}\right) \tag{4}$$

$$A^{\ell,j} = \gamma\left(\frac{\left(X^{\ell-1}W_Q^{\ell,j}\right)\left(X^{\ell-1}W_K^{\ell,j}\right)^T}{\sqrt{d/H}} + M^{\ell,j}\right) \tag{5}$$

where $X^\ell \in \mathbb{R}^{N\times d}$ is a matrix with all token representations at layer ℓ, γ is a row-wise softmax normalization, $A^{\ell,j} \in \mathbb{R}^{N\times N}$ encodes the weights computed by the j-th attention head at layer ℓ, and $M^{\ell,j}$ is a mask for $A^{\ell,j}$. In auto-regressive LMs, $A^{\ell,j}$ is masked to a lower triangular matrix, as each position can only attend to preceding positions (i.e. $M_{rc}^{\ell,j} = -\infty\ \forall c > r$ and zero otherwise). Importantly, the cell $A_{rc}^{\ell,j}$ can viewed as a weighted edge from the r-th to the c-th hidden representations at layer $\ell - 1$.

MLP Sublayers Every MLP sublayer computes a *local* update for each representation:

$$\mathbf{m}_i^\ell = W_F^\ell\, \sigma\left(W_I^\ell\left(\mathbf{a}_i^\ell + \mathbf{x}_i^{\ell-1}\right)\right) \tag{6}$$

where $W_I^\ell \in \mathbb{R}^{d_i\times d}$ and $W_F^\ell \in \mathbb{R}^{d\times d_i}$ are parameter matrices with inner-dimension d_i, and σ is a nonlinear activation function. Recent works has shown that transformer MLP sublayers can be cast as key-value memories (Geva et al., 2021) that store factual knowledge (Dai et al., 2022; Meng et al., 2022a).

3 Experimental Setup

We focus on the task of factual open-domain questions, where a model needs to predict an attribute a of a given subject-relation pair (s, r). A triplet (s, r, a) is typically expressed in a question-answering format (e.g. *"What instrument did Elvis Presley play?"*) or as a fill-in-the-blank query (e.g. *"Elvis Presley played the ____"*). While LMs often succeed at predicting the correct attribute for such queries (Roberts et al., 2020; Petroni et al., 2019), it is unknown how attributes are extracted internally.

For a factual query q that expresses the subject s and relation r of a triplet (s, r, a), let $t = (t_1, ..., t_N)$ be the representation of q as a sequence of tokens, based on some LM. We refer by the *subject tokens* to the sub-sequence of t that corresponds to s, and by the *subject positions* to the positions of the subject tokens in t. The non-subject tokens in q express the relation r.

Data We use queries from COUNTERFACT (Meng et al., 2022a). For a given model, we extract a random sample of queries for which the model predicts the correct attribute. In the rest of the paper, we refer to the token predicted by the model for a given query q as the attribute a, even though it could be a sub-word and thus only the prefix of the attribute name (e.g. `Wash` for "Washington").

Models We analyze two auto-regressive decoder-only GPT LMs with different layouts: GPT-2 (Radford et al., 2019) ($L = 48$, 1.5B parameters) and GPT-J (Wang and Komatsuzaki, 2021) ($L = 28$, 6B parameters). Both models use a vocabulary with ~50K tokens. Also, GPT-J employs parallel MHSA and MLP sublayers, where the output of the ℓ-th MLP sublayer for the i-th representation depends on $\mathbf{x}_i^{\ell-1}$ rather than on $\mathbf{a}_i^\ell + \mathbf{x}_i^{\ell-1}$ (see Eq. 6). We follow the procedure by Meng et al. (2022a) to create a data sample for each model, resulting in 1,209 queries for GPT-2 and 1,199 for GPT-J.

4 Overview: Experiments & Findings

We start by introducing our attention blocking method and apply it to identify critical information flow points in factual predictions (§5) – one from the relation, followed by another from the subject. Then, we analyze the evolution of the subject representation in the layers preceding this critical point (§6), and find that it goes through an enrichment process driven by the MLP sublayers, to encode many subject-related attributes. Last, we investigate how and where the right attribute is extracted from this representation (§7), and discover that this is typically done by the upper MHSA sublayers, via attention heads that often encode a subject-attribute mapping in their parameters.